# betaMC: Staging

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## 1 Monte Carlo Simulation

```
# Fit the regression model
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
# Generate the sampling distribution of parameter estimates
mc <- MC(object, R = 20000, type = "mvn", seed = 42)
```

## 2 Standardized Regression Slopes

```
out <- BetaMC(mc)
# Methods --------------------------------------------------------
print(out)
```

```
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0762 20000 0.2361 0.2901 0.3370 0.6365 0.6806 0.7274
#> PCTGRT  0.3915 0.0772 20000 0.1331 0.1904 0.2356 0.5387 0.5835 0.6315
#> PCTSUPP 0.2632 0.0750 20000 0.0264 0.0749 0.1180 0.4116 0.4624 0.5287
```

```
summary(out)
```

```
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0762 20000 0.2361 0.2901 0.3370 0.6365 0.6806 0.7274
#> PCTGRT  0.3915 0.0772 20000 0.1331 0.1904 0.2356 0.5387 0.5835 0.6315
#> PCTSUPP 0.2632 0.0750 20000 0.0264 0.0749 0.1180 0.4116 0.4624 0.5287
```

```r
coef(out)
```

```
#>    NARTIC    PCTGRT   PCTSUPP 
#> 0.4951451 0.3914887 0.2632477
```

```r
vcov(out)
```

```
#>               NARTIC       PCTGRT      PCTSUPP
#> NARTIC   0.005813246 -0.003327552 -0.002189672
#> PCTGRT  -0.003327552  0.005965054 -0.001733688
#> PCTSUPP -0.002189672 -0.001733688  0.005621350
```

```r
confint(out)
```

```
#>             2.5 %    97.5 %
#> NARTIC  0.3369847 0.6365108
#> PCTGRT  0.2356129 0.5387022
#> PCTSUPP 0.1179584 0.4115677
```

# 3 Multiple Correlation

```r
out <- RSqMC(mc)
# Methods --------------------------------------------------------
print(out)
```

```
#> Call:
#> RSqMC(object = mc)
#> 
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0563 20000 0.5253 0.5927 0.6616 0.8811 0.9026 0.9275
#> adj 0.7906 0.0603 20000 0.4914 0.5636 0.6374 0.8726 0.8956 0.9223
```

```r
summary(out)
```

```
#> Call:
#> RSqMC(object = mc)
#> 
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0563 20000 0.5253 0.5927 0.6616 0.8811 0.9026 0.9275
#> adj 0.7906 0.0603 20000 0.4914 0.5636 0.6374 0.8726 0.8956 0.9223
```

```r
coef(out)

#>       rsq       adj 
#> 0.8045263 0.7905638

vcov(out)

#>             rsq         adj
#> rsq 0.003167170 0.003393397
#> adj 0.003393397 0.003635782

confint(out)

#>         2.5 %    97.5 %
#> rsq 0.6615936 0.8811200
#> adj 0.6374217 0.8726286
```

# 4 Semipartial Correlation

```r
out <- SCorMC(mc)
# Methods ---------------------------------------------------------
print(out)

#> Call:
#> SCorMC(object = mc)
#> 
#> Semipartial correlations
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0778 20000 0.1722 0.2265 0.2692 0.5719 0.6262 0.7016
#> PCTGRT  0.3430 0.0745 20000 0.1140 0.1549 0.1912 0.4856 0.5328 0.5902
#> PCTSUPP 0.2385 0.0700 20000 0.0220 0.0646 0.1018 0.3774 0.4274 0.4954

summary(out)

#> Call:
#> SCorMC(object = mc)
#> 
#> Semipartial correlations
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0778 20000 0.1722 0.2265 0.2692 0.5719 0.6262 0.7016
#> PCTGRT  0.3430 0.0745 20000 0.1140 0.1549 0.1912 0.4856 0.5328 0.5902
#> PCTSUPP 0.2385 0.0700 20000 0.0220 0.0646 0.1018 0.3774 0.4274 0.4954
```

```r
coef(out)
```

```
#>   NARTIC    PCTGRT   PCTSUPP
#> 0.4311525 0.3430075 0.2384789
```

```r
vcov(out)
```

```
#>               NARTIC        PCTGRT       PCTSUPP
#> NARTIC   0.006047170 -0.0012581274 -0.0009426070
#> PCTGRT  -0.001258127  0.0055564117 -0.0007626883
#> PCTSUPP -0.000942607 -0.0007626883  0.0048961207
```

```r
confint(out)
```

```
#>             2.5 %    97.5 %
#> NARTIC  0.2692038 0.5718566
#> PCTGRT  0.1911913 0.4856258
#> PCTSUPP 0.1017820 0.3773570
```

# 5 Improvement in R-Squared

```r
out <- DeltaRSqMC(mc)
# Methods ---------------------------------------------------------
print(out)
```

```
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.1859 0.0662 20000 0.0297 0.0513 0.0725 0.3270 0.3921 0.4922
#> PCTGRT  0.1177 0.0511 20000 0.0130 0.0240 0.0366 0.2358 0.2838 0.3483
#> PCTSUPP 0.0569 0.0343 20000 0.0005 0.0042 0.0104 0.1424 0.1827 0.2454
```

```r
summary(out)
```

```
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.1859 0.0662 20000 0.0297 0.0513 0.0725 0.3270 0.3921 0.4922
#> PCTGRT  0.1177 0.0511 20000 0.0130 0.0240 0.0366 0.2358 0.2838 0.3483
#> PCTSUPP 0.0569 0.0343 20000 0.0005 0.0042 0.0104 0.1424 0.1827 0.2454
```

```r
coef(out)
```

```
#>    NARTIC    PCTGRT   PCTSUPP 
#> 0.1858925 0.1176542 0.0568722
```

```r
vcov(out)
```

```
#>                NARTIC        PCTGRT       PCTSUPP
#> NARTIC   0.0043829298 -0.0007249412 -0.0003689302
#> PCTGRT  -0.0007249412  0.0026113477 -0.0002312324
#> PCTSUPP -0.0003689302 -0.0002312324  0.0011748195
```

```r
confint(out)
```

```
#>              2.5 %    97.5 %
#> NARTIC  0.07247067 0.3270199
#> PCTGRT  0.03655411 0.2358325
#> PCTSUPP 0.01035958 0.1423983
```

## 6 Squared Partial Correlation

```r
out <- PCorMC(mc)
# Methods -------------------------------------------------------
print(out)
```

```
#> Call:
#> PCorMC(object = mc)
#> 
#> Squared partial correlations
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.1057 20000 0.1146 0.1804 0.2429 0.6539 0.7100 0.7804
#> PCTGRT  0.3757 0.1079 20000 0.0534 0.0970 0.1436 0.5659 0.6314 0.7051
#> PCTSUPP 0.2254 0.1001 20000 0.0016 0.0187 0.0460 0.4295 0.5111 0.6142
```

```r
summary(out)
```


```
#> Call:
#> PCorMC(object = mc)
#> 
#> Squared partial correlations
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.1057 20000 0.1146 0.1804 0.2429 0.6539 0.7100 0.7804
#> PCTGRT  0.3757 0.1079 20000 0.0534 0.0970 0.1436 0.5659 0.6314 0.7051
#> PCTSUPP 0.2254 0.1001 20000 0.0016 0.0187 0.0460 0.4295 0.5111 0.6142
```

```r
coef(out)
```

```
#>    NARTIC    PCTGRT   PCTSUPP
#> 0.4874382 0.3757383 0.2253739
```

```r
vcov(out)
```

```
#>               NARTIC       PCTGRT      PCTSUPP
#> NARTIC  0.0111660142 0.0006140081 0.0003672582
#> PCTGRT  0.0006140081 0.0116449785 0.0001529567
#> PCTSUPP 0.0003672582 0.0001529567 0.0100205185
```

```r
confint(out)
```

```
#>              2.5 %    97.5 %
#> NARTIC  0.24287903 0.6539426
#> PCTGRT  0.14355226 0.5658939
#> PCTSUPP 0.04600206 0.4294976
```

# 7 Differences of Standardized Slopes

```r
out <- DiffBetaMC(mc)
# Methods --------------------------------------------------------
print(out)
```

```
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                    est     se     R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1037 0.1358 20000 -0.3321 -0.2463 -0.1645 0.3659 0.4490 0.5284
#> NARTIC-PCTSUPP 0.2319 0.1258 20000 -0.1953 -0.1137 -0.0216 0.4687 0.5409 0.6060
#> PCTGRT-PCTSUPP 0.1282 0.1227 20000 -0.2731 -0.1934 -0.1181 0.3625 0.4395 0.5274
```

```r
summary(out)
```

```
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                    est     se     R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1037 0.1358 20000 -0.3321 -0.2463 -0.1645 0.3659 0.4490 0.5284
#> NARTIC-PCTSUPP 0.2319 0.1258 20000 -0.1953 -0.1137 -0.0216 0.4687 0.5409 0.6060
#> PCTGRT-PCTSUPP 0.1282 0.1227 20000 -0.2731 -0.1934 -0.1181 0.3625 0.4395 0.5274
```

```r
coef(out)

#>  NARTIC-PCTGRT NARTIC-PCTSUPP PCTGRT-PCTSUPP
#>      0.1036564      0.2318974      0.1282410

vcov(out)

#>                NARTIC-PCTGRT NARTIC-PCTSUPP PCTGRT-PCTSUPP
#> NARTIC-PCTGRT    0.018433405    0.009596782   -0.008836623
#> NARTIC-PCTSUPP   0.009596782    0.015813941    0.006217159
#> PCTGRT-PCTSUPP  -0.008836623    0.006217159    0.015053781

confint(out)

#>                      2.5 %    97.5 %
#> NARTIC-PCTGRT  -0.16450381 0.3659226
#> NARTIC-PCTSUPP -0.02163246 0.4687397
#> PCTGRT-PCTSUPP -0.11805414 0.3624568
```

# 8 Monte Carlo Simulation - Multiple Imputation

```r
nas1982_missing <- mice::ampute(nas1982, mech = "MCAR")$amp
# Fit the regression model
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982_missing)
# Generate the sampling distribution of parameter estimates
mi <- mice::mice(nas1982_missing, m = 100, seed = 42, print = FALSE)
mc <- MCMI(object, mi = mi, R = 20000, type = "mvn",
           seed = 42)
```

# 9 Standardized Regression Slopes

```r
out <- BetaMC(mc)
# Methods --------------------------------------------------------
print(out)

#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se     R   0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.5222 0.0799 20000  0.2480 0.3029 0.3556 0.6673 0.7081 0.7682
```

```
#> PCTGRT  0.3797 0.0807 20000  0.1037 0.1745 0.2195 0.5373 0.5819 0.6306
#> PCTSUPP 0.2432 0.0829 20000 -0.0189 0.0360 0.0820 0.4081 0.4615 0.5381

summary(out)

#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se     R   0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.5222 0.0799 20000  0.2480 0.3029 0.3556 0.6673 0.7081 0.7682
#> PCTGRT  0.3797 0.0807 20000  0.1037 0.1745 0.2195 0.5373 0.5819 0.6306
#> PCTSUPP 0.2432 0.0829 20000 -0.0189 0.0360 0.0820 0.4081 0.4615 0.5381

coef(out)

#>    NARTIC    PCTGRT   PCTSUPP
#> 0.5222438 0.3796524 0.2431948

vcov(out)

#>               NARTIC       PCTGRT      PCTSUPP
#> NARTIC   0.006377888 -0.003527483 -0.002475092
#> PCTGRT  -0.003527483  0.006509939 -0.002009889
#> PCTSUPP -0.002475092 -0.002009889  0.006873409

confint(out)

#>              2.5 %    97.5 %
#> NARTIC  0.35558830 0.6672895
#> PCTGRT  0.21950783 0.5372601
#> PCTSUPP 0.08200066 0.4080524
```

# 10 Multiple Correlation

```
out <- RSqMC(mc)
# Methods ---------------------------------------------------------
print(out)

#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
```

```
#>        est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8029 0.0619 20000 0.5064 0.5906 0.6494 0.8914 0.9209 0.9476
#> adj 0.7889 0.0686 20000 0.4535 0.5468 0.6119 0.8798 0.9124 0.9420
```

```r
summary(out)
```

```
#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8029 0.0619 20000 0.5064 0.5906 0.6494 0.8914 0.9209 0.9476
#> adj 0.7889 0.0686 20000 0.4535 0.5468 0.6119 0.8798 0.9124 0.9420
```

```r
coef(out)
```

```
#>       rsq       adj
#> 0.8029376 0.7888617
```

```r
vcov(out)
```

```
#>             rsq         adj
#> rsq 0.003833959 0.004244740
#> adj 0.004244740 0.004699534
```

```r
confint(out)
```

```
#>         2.5 %    97.5 %
#> rsq 0.6494141 0.8914298
#> adj 0.6118513 0.8797973
```

# 11 Semipartial Correlation

```r
out <- SCorMC(mc)
# Methods --------------------------------------------------------
print(out)
```

```
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>            est     se     R   0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4601 0.0834 20000  0.1655 0.2350 0.2836 0.6101 0.6620 0.7369
```

```
#> PCTGRT  0.3312 0.0771 20000  0.0790 0.1377 0.1776 0.4812 0.5307 0.5883
#> PCTSUPP 0.2183 0.0766 20000 -0.0159 0.0294 0.0691 0.3696 0.4247 0.4975
```

```r
summary(out)
```

```
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>            est     se     R   0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4601 0.0834 20000  0.1655 0.2350 0.2836 0.6101 0.6620 0.7369
#> PCTGRT  0.3312 0.0771 20000  0.0790 0.1377 0.1776 0.4812 0.5307 0.5883
#> PCTSUPP 0.2183 0.0766 20000 -0.0159 0.0294 0.0691 0.3696 0.4247 0.4975
```

```r
coef(out)
```

```
#>    NARTIC    PCTGRT   PCTSUPP
#> 0.4601125 0.3311773 0.2183476
```

```r
vcov(out)
```

```
#>               NARTIC       PCTGRT      PCTSUPP
#> NARTIC   0.006952654 -0.001396679 -0.001045336
#> PCTGRT  -0.001396679  0.005938199 -0.000913114
#> PCTSUPP -0.001045336 -0.000913114  0.005870865
```

```r
confint(out)
```

```
#>              2.5 %    97.5 %
#> NARTIC  0.28356805 0.6101212
#> PCTGRT  0.17761296 0.4811563
#> PCTSUPP 0.06913635 0.3696167
```

# 12 Improvement in R-Squared

```r
out <- DeltaRSqMC(mc)
# Methods --------------------------------------------------------
print(out)
```

```
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
```

```
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.2117 0.0750 20000 0.0274 0.0552 0.0804 0.3722 0.4382 0.5431
#> PCTGRT  0.1097 0.0512 20000 0.0062 0.0190 0.0315 0.2315 0.2816 0.3461
#> PCTSUPP 0.0477 0.0348 20000 0.0000 0.0009 0.0048 0.1366 0.1804 0.2475
```

```
summary(out)
```

```
#> Call:
#> DeltaRSqMC(object = mc)
#> 
#> Improvement in R-squared
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.2117 0.0750 20000 0.0274 0.0552 0.0804 0.3722 0.4382 0.5431
#> PCTGRT  0.1097 0.0512 20000 0.0062 0.0190 0.0315 0.2315 0.2816 0.3461
#> PCTSUPP 0.0477 0.0348 20000 0.0000 0.0009 0.0048 0.1366 0.1804 0.2475
```

```
coef(out)
```

```
#>     NARTIC     PCTGRT    PCTSUPP 
#> 0.21170355 0.10967840 0.04767569
```

```
vcov(out)
```

```
#>                NARTIC        PCTGRT       PCTSUPP
#> NARTIC   0.0056235675 -0.0008247462 -0.0004056533
#> PCTGRT  -0.0008247462  0.0026233718 -0.0002439773
#> PCTSUPP -0.0004056533 -0.0002439773  0.0012077140
```

```
confint(out)
```

```
#>               2.5 %    97.5 %
#> NARTIC  0.080410838 0.3722479
#> PCTGRT  0.031546364 0.2315113
#> PCTSUPP 0.004779835 0.1366165
```

# 13 Squared Partial Correlation

```
out <- PCorMC(mc)
# Methods -----------------------------------------------------------
print(out)
```

```
#> Call:
#> PCorMC(object = mc)
#> 
```

```
#> Squared partial correlations
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.5188 0.1148 20000 0.1108 0.1855 0.2532 0.6997 0.7646 0.8282
#> PCTGRT  0.3592 0.1154 20000 0.0327 0.0789 0.1249 0.5717 0.6529 0.7541
#> PCTSUPP 0.1989 0.1095 20000 0.0001 0.0044 0.0216 0.4374 0.5282 0.6262
```

```r
summary(out)
```

```
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>            est     se     R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.5188 0.1148 20000 0.1108 0.1855 0.2532 0.6997 0.7646 0.8282
#> PCTGRT  0.3592 0.1154 20000 0.0327 0.0789 0.1249 0.5717 0.6529 0.7541
#> PCTSUPP 0.1989 0.1095 20000 0.0001 0.0044 0.0216 0.4374 0.5282 0.6262
```

```r
coef(out)
```

```
#>    NARTIC    PCTGRT   PCTSUPP
#> 0.5187942 0.3591718 0.1989111
```

```r
vcov(out)
```

```
#>              NARTIC       PCTGRT      PCTSUPP
#> NARTIC  0.013180180 0.0017380785 0.0015036079
#> PCTGRT  0.001738078 0.0133056896 0.0007733401
#> PCTSUPP 0.001503608 0.0007733401 0.0120007991
```

```r
confint(out)
```

```
#>              2.5 %    97.5 %
#> NARTIC  0.25315794 0.6997445
#> PCTGRT  0.12493163 0.5716581
#> PCTSUPP 0.02160196 0.4374294
```

# 14 Differences of Standardized Slopes

```r
out <- DiffBetaMC(mc)
# Methods --------------------------------------------------------
print(out)
```

```
#> Call:
```

```
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                   est     se     R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1426 0.1412 20000 -0.3148 -0.2252 -0.1404 0.4107 0.4915 0.5825
#> NARTIC-PCTSUPP 0.2790 0.1349 20000 -0.1926 -0.0856  0.0019 0.5299 0.6047 0.6728
#> PCTGRT-PCTSUPP 0.1365 0.1319 20000 -0.2912 -0.2064 -0.1270 0.3922 0.4724 0.5646
```

```
summary(out)
```

```
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                   est     se     R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1426 0.1412 20000 -0.3148 -0.2252 -0.1404 0.4107 0.4915 0.5825
#> NARTIC-PCTSUPP 0.2790 0.1349 20000 -0.1926 -0.0856  0.0019 0.5299 0.6047 0.6728
#> PCTGRT-PCTSUPP 0.1365 0.1319 20000 -0.2912 -0.2064 -0.1270 0.3922 0.4724 0.5646
```

```
coef(out)
```

```
#>  NARTIC-PCTGRT NARTIC-PCTSUPP PCTGRT-PCTSUPP
#>      0.1425914      0.2790490      0.1364576
```

```
vcov(out)
```

```
#>                NARTIC-PCTGRT NARTIC-PCTSUPP PCTGRT-PCTSUPP
#> NARTIC-PCTGRT     0.01994279    0.010370574   -0.009572220
#> NARTIC-PCTSUPP    0.01037057    0.018201480    0.007830906
#> PCTGRT-PCTSUPP   -0.00957222    0.007830906    0.017403126
```

```
confint(out)
```

```
#>                       2.5 %    97.5 %
#> NARTIC-PCTGRT  -0.140425212 0.4107107
#> NARTIC-PCTSUPP  0.001942493 0.5298746
#> PCTGRT-PCTSUPP -0.127045378 0.3922386
```

# References


R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/